they are free of Cu, P, Ni, Mn and S impurities). Significant radiation damage in the reactor vessels due to impurities of Ni and Mn has been observed in both BWR and VVER reactors. Stainless steel (e.g. of the 18% Cr, 8% Ni type) has been found to become brittle upon irradiation due to the formation of microscopic helium bubbles, probably due to n,$\alpha$ reactions in $^{54}$Fe and impurities of light elements (N, B, etc.). This behaviour is accentuated for metallic uranium in reactors because of the formation of fission products, some of which are gases. As a result of this radiation effect it is not possible to use uranium metal in modern power reactors, where high radiation doses are accumulated in a very short time. The fuel elements for power reactors are therefore made of non-metallic uranium compounds. However, in some early fast reactors, e.g. EBR II, metallic uranium fuel was used successfully by increasing the gap between the fuel and the can and filling it with sodium.

The displaced atoms may return to their original lattice positions through diffusion if they are not trapped in energy wells requiring some activation energy for release. Such energy can be provided by heating or by irradiation with electrons or $\gamma$-rays (these do not cause new displacements). This "healing" of particle radiation damage is commonly referred to as annealing. The *thermal annealing rate* increases with temperature as does *radiation annealing* with radiation dose. Doses in the 10 kGy range are usually required for appreciable effect.

## 8.5. INORGANIC NON-METALLIC COMPOUNDS

The time for a high energy particle to pass by an atom is $\leq 10^{-16}$ s. In this time the atom may become excited and/or ionized, but it does not change position (the *Franck-Condon principle*) provided there is no direct collision. The excited atoms are de-excited through the emission of *fluorescence radiation*, usually within $10^{-8}$ s. The ionization can result in simple trapping of the electrons and production of "electron holes" in the lattice, especially at impurity sites. The local excess (or deficiency) of charge produced in this way leads to electronic states with absorption bands in the visible and ultraviolet regions of the spectrum. For example, irradiation of LiCl results in a change of the colour of the compound from white to yellow. Similarly, LiF becomes black, KCl blue, etc. The irradiation of ionic crystals also leads to changes in other physical properties much as conductivity, hardness, strength, etc. Frequently, heating (sometimes room temperature is enough) returns the properties and colour to the normal state (or close to it) accompanied by the emission of light; this forms the basis for a radiation dose measurement technique named "thermoluminescence dosimetry" (§8.10).

Following a collision between a heavy particle (n, p, etc.) and an absorber atom in a crystalline material the recoiling ion produces lattice vacancies and, upon stopping, may occupy a non-equilibrium interstitial position (Fig. 8.3). The localized dissipation of energy can result in lattice oscillations, terminating in some reorientation of the local

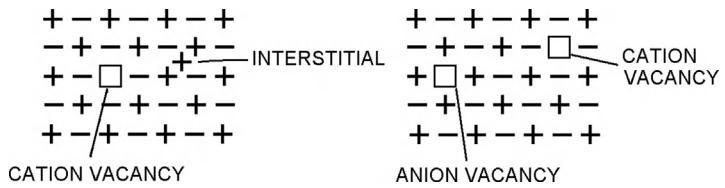



**Figure 8.3** Irradiated NaCl type crystal showing negative and positive ion vacancies.

regions in the crystal lattice. These crystal defects increase the energy content of the crystal. Semiconductors, where the concentration of charge carriers is very small, have their conductivity reduced by introduction of lattice defects during irradiation. The production of interstitial atoms makes the graphite moderator in nuclear reactors stronger, harder, and more brittle. Since these dislocated atoms are more energetic than the atoms in the lattice, the dislocations lead to an energy storage in the material (the *Wigner effect*), which can become quite significant. For reactor graphite at 30°C this frequently reaches values as high as 2000 kJ/kg for fluencies of $2 \times 10^{21}$ n/cm$^2$. At room temperature the interstitial atoms return to their normal positions very slowly (annealing), but this rate is quite temperature dependent. If the elimination of the interstitial atoms occurs too rapidly, the release of the Wigner energy can cause the material to heat to the ignition point. This was the origin of a fire which occurred in a graphite moderated reactor in England in 1957, in which considerable amounts of radioactive fission products were released into the environment.

Inorganic substances exposed to high fluencies of neutron and γ-radiation in nuclear reactors are found to experience decomposition. Thus:

$$KNO_3 \ \sim\sim\sim\sim\rightarrow \ KNO_2 + \tfrac{1}{2}O_2 \tag{8.15}$$

The notation $\sim\sim\sim\sim\rightarrow$ symbolizes a radiation induced reaction. At high fluxes the oxygen pressure in the $KNO_3$ causes the crystal to shatter. However, some crystals are remarkably stable (although they may become coloured), e.g. $Li_2SO_4$, $K_2SO_4$, $KCrO_4$, and $CaCO_3$.

Theory has not been developed sufficiently to allow quantitative calculation of the radiation sensitivity of compounds. Usually covalent binary compounds are highly radiation resistant. Two examples are $UO_2$ and UC, whose insensitivity to radiation has led to their use as reactor fuels. In the fuel elements of commercial water cooled reactors $UO_2$ is used in the form of small, sintered pellets about 1 cm$^3$ in volume. Because of the build-up of pressure from fission gases, these pellets crack at high fluxes ($\geq 10^{22}$ n/cm$^2$); Figure 8.4. Another binary compound, $CO_2$, which is used as a coolant in some older reactors, decomposes by irradiation to graphite and polymeric species.

In mixtures of inorganic compounds many unexpected and even undesirable reactions may occur. For example, radiolysis of liquid air (often used in radiation

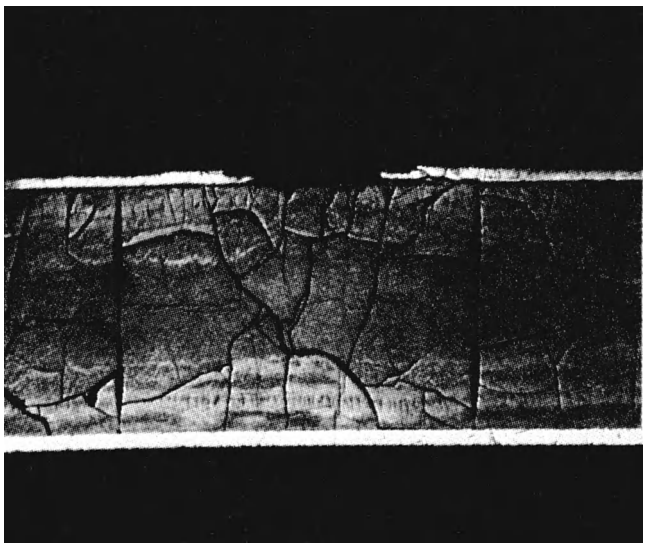


**Figure 8.4** Zircaloy canned $UO_2$ fuel after exposure to very high power (high dose and heat) in a test reactor. The pellets have fractured and the canning has been penetrated. (SKI report)

research) yields ozone, while radiation of humid air yields $HNO_3$. One of the first observations of radiation-induced changes was the darkening of glass. Glass often contains iron, manganese, and other metals that can exist in several oxidation states with different colours. As a result of the irradiation, the oxidation state can change, resulting in change in colour. Dislocations as well as trapped electrons also contribute to the colour changes in glass. In chemical and metallurgical work with highly active substances it is desirable to observe the experiment through a thick glass window, which provides protection from the radiation. In order to avoid the colouring of the glass, a small amount (1−2%) of an element which can act as an electron trap is added, e.g. $CeO_2$, which acts by the reaction $Ce^{4+} + e^- \rightarrow Ce^{3+}$. After an exposure of $10^4$ Gy the transmission to light of ordinary glass had been reduced to 44%, while for a $CeO_2$ protected glass it was still 89%.

Glass is very resistant to radiation damage because it is a non-crystalline "solid" liquid. Therefore, in such a material, it is not possible to speak of dislocations: the random structure of the glass allows it to include foreign species throughout the sample. This is why glass has been intensely studied, including full scale tests, as the matrix ("container") for high active waste (HAW) consisting of fission products and actinides.

## 8.6. WATER

The consequences of ionization and excitation depend on the physical state and the molecular composition of the irradiated material. In this section we introduce chemical phenomena into the description using the radiolysis of water as an example.